

Total Presentation Time is 4 min

RAMSES



R. Teyssier, J. Palafoutas, R. Geda, J. Sunseri
Y. Pan, J-Y Lee, W. Groger
Princeton University

E. Moseley
Stanford University

B. Loring, G. Barnier
NVidia



RAMSES Main Routines

HYDRO

- Godunov MUSCL-Hancock HLLC Riemann solver
- 2 kernels (cube and paper)
- nsubgrid = 1, 2 or 4

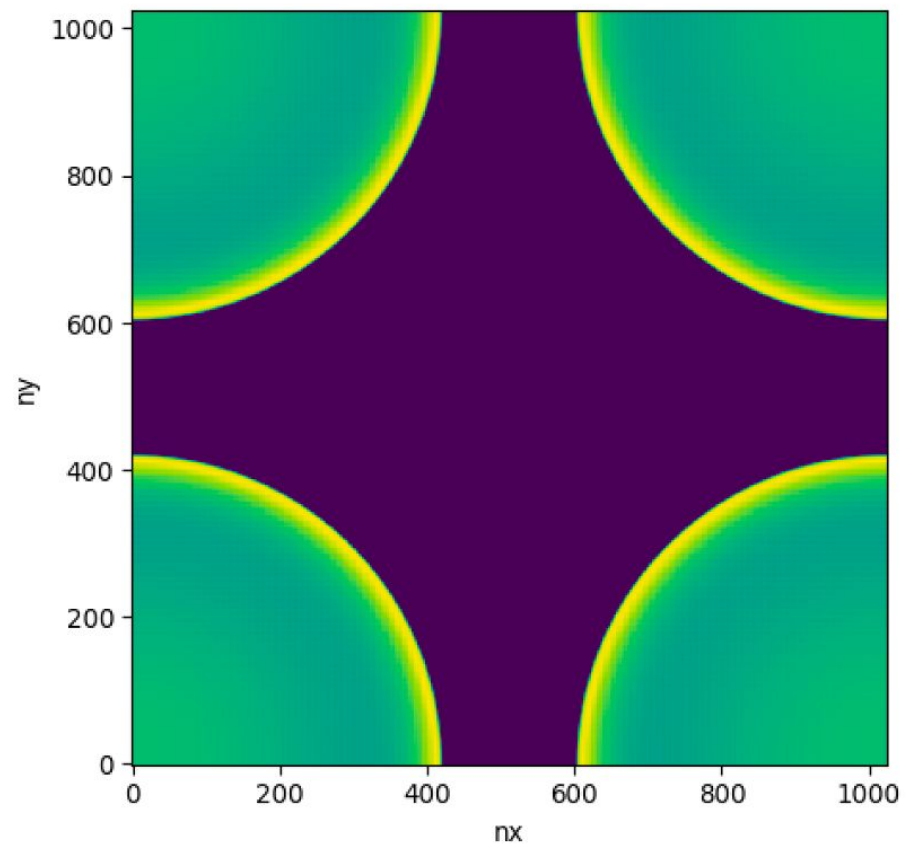
GRAVITY

- Multigrid (MG) Poisson solver
- Hierarchical multi-resolution Gauss-Seidel iteration (V-cycle)

PARTICLES

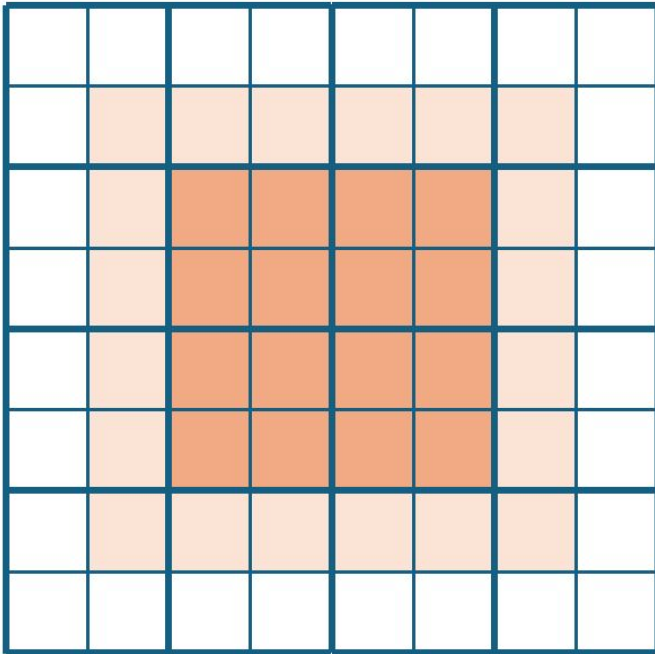
- Mass deposition
- Kick-Drift-Kick pusher

HYDRO SOLVER

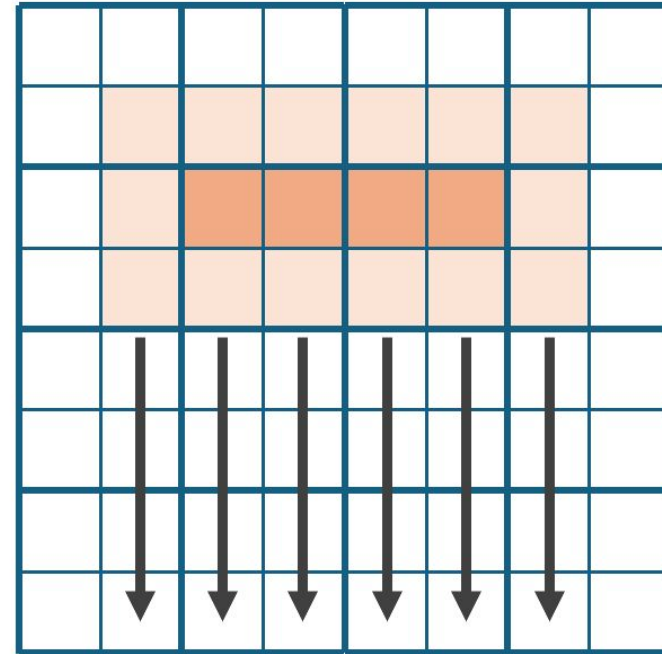


Optimizing shared memory with paper kernel

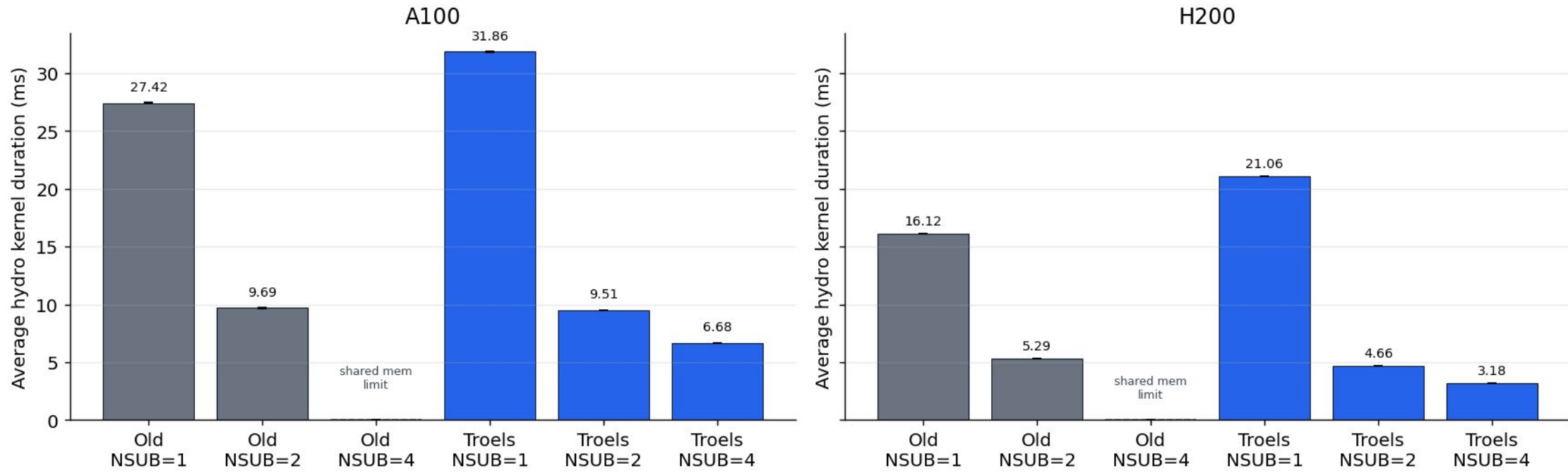
Caddy kernel, 216 cells



Paper kernel, 108 cells swept downward



Sedov3D Unigrid level=8: Average Hydro Kernel Call Time
Single precision, nsys CUDA kernel duration





Paper kernel, nsubgrid=4

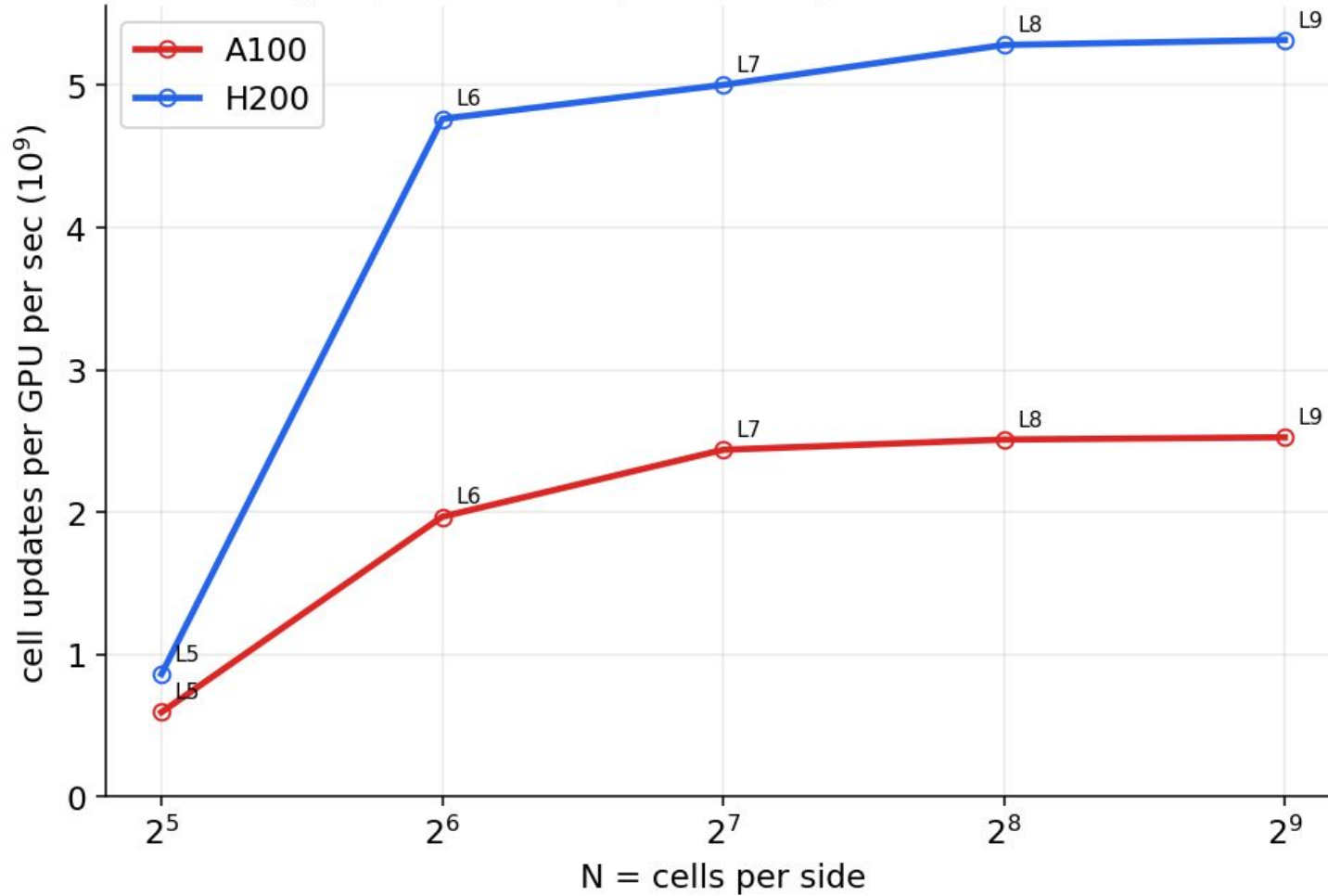


Caddy kernel, nsubgrid=2

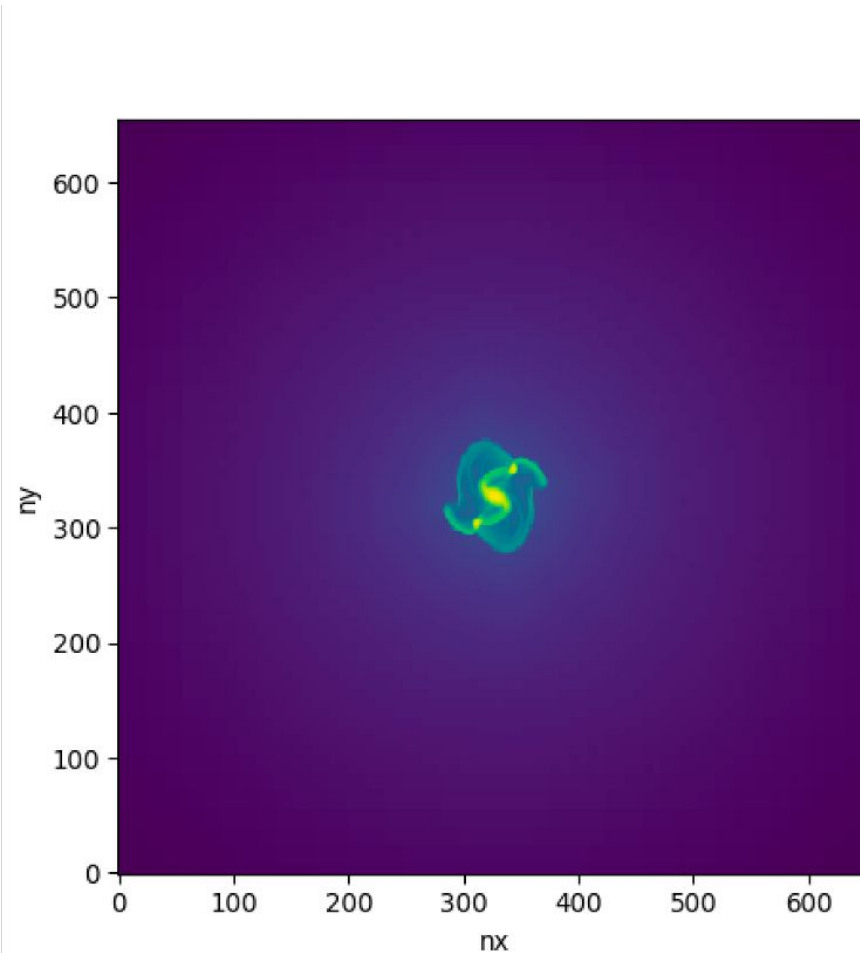


Caddy kernel, nsubgrid=1

Sedov3D Unigrid: Troels Hydro NSUBGRID=4
Single precision, nsys CUDA hydro-kernel duration



GRAVITY SOLVER

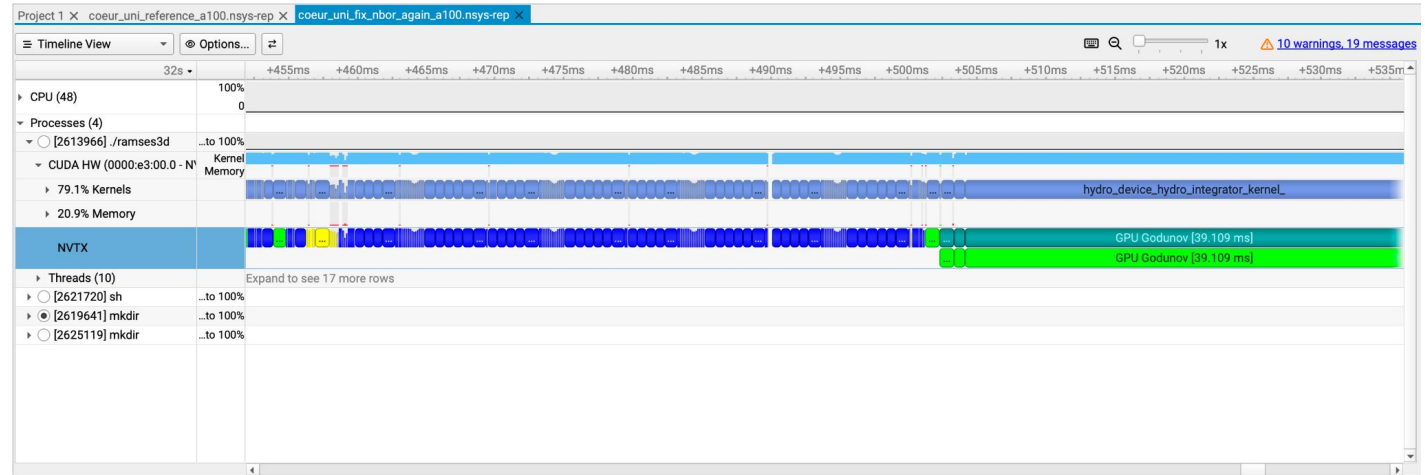
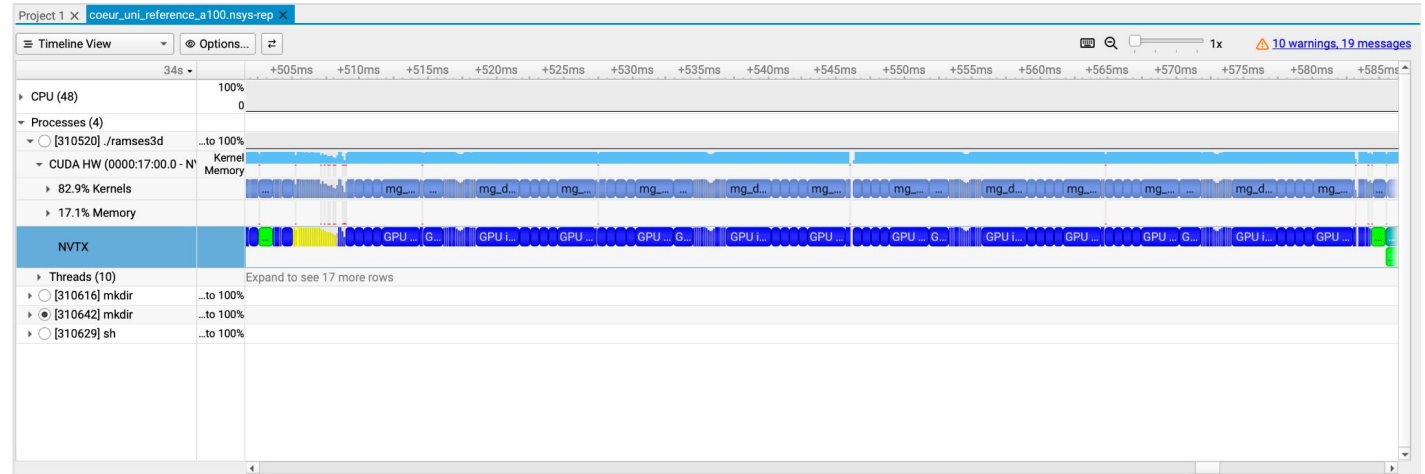


NSight Systems Output and Easy Optimizations

Minimized D2H and
H2D memory copies

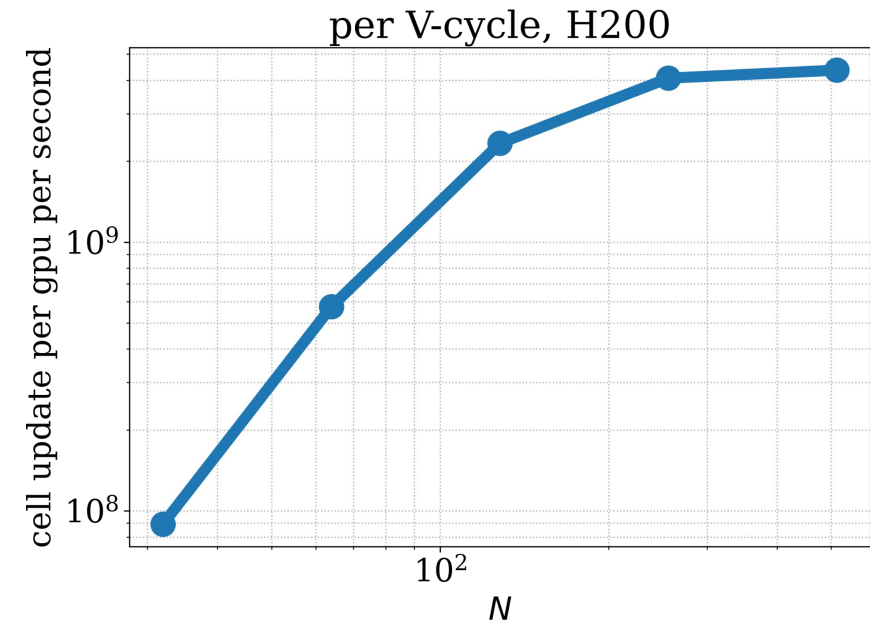
Fused kernels

Minimized number of
main GPU memory
access

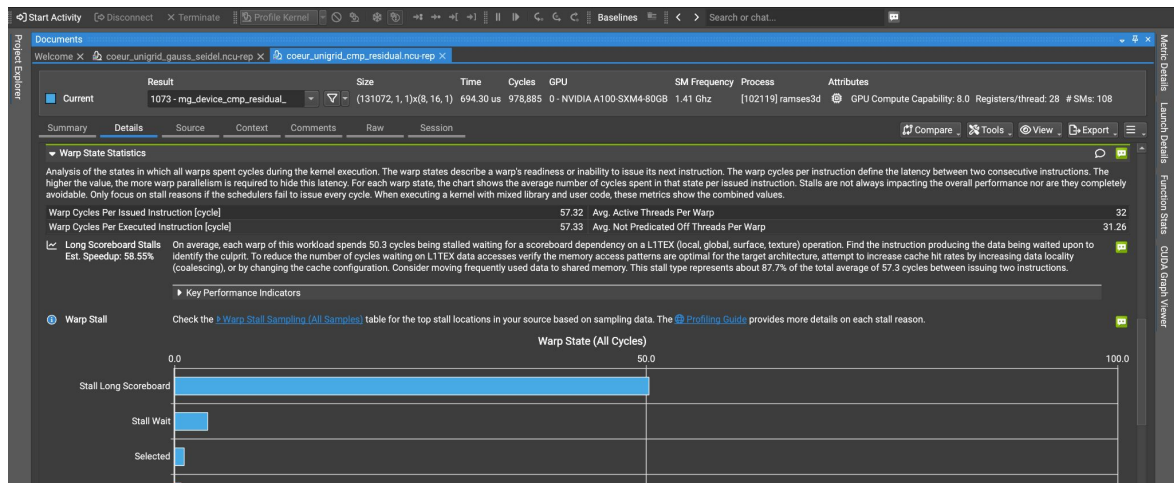
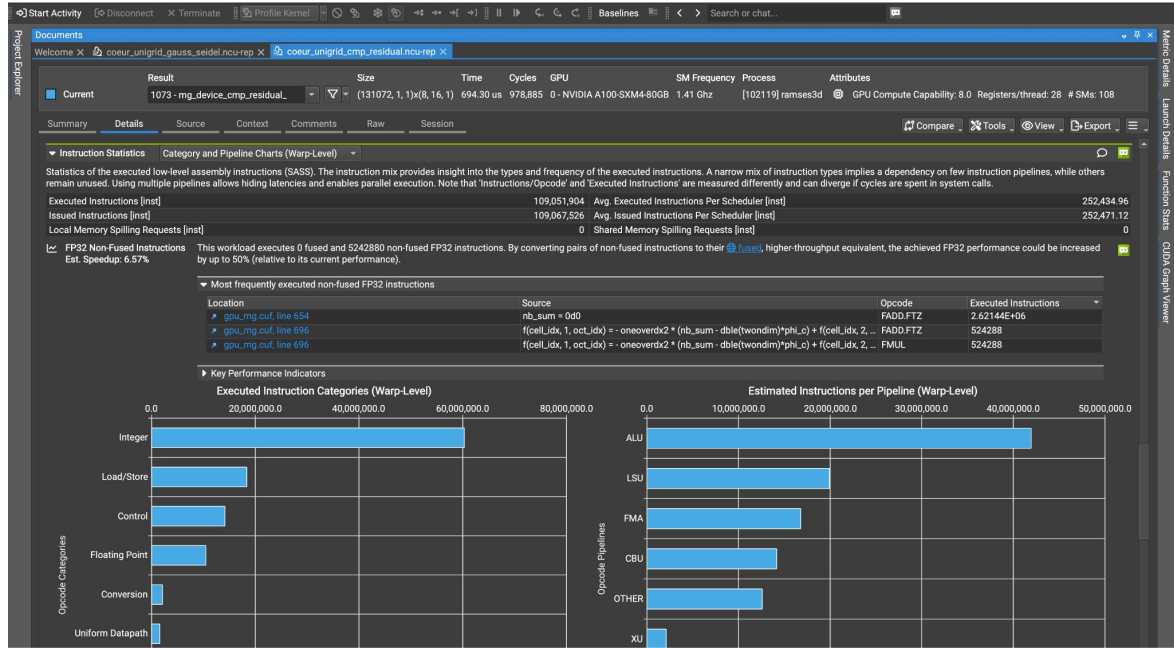
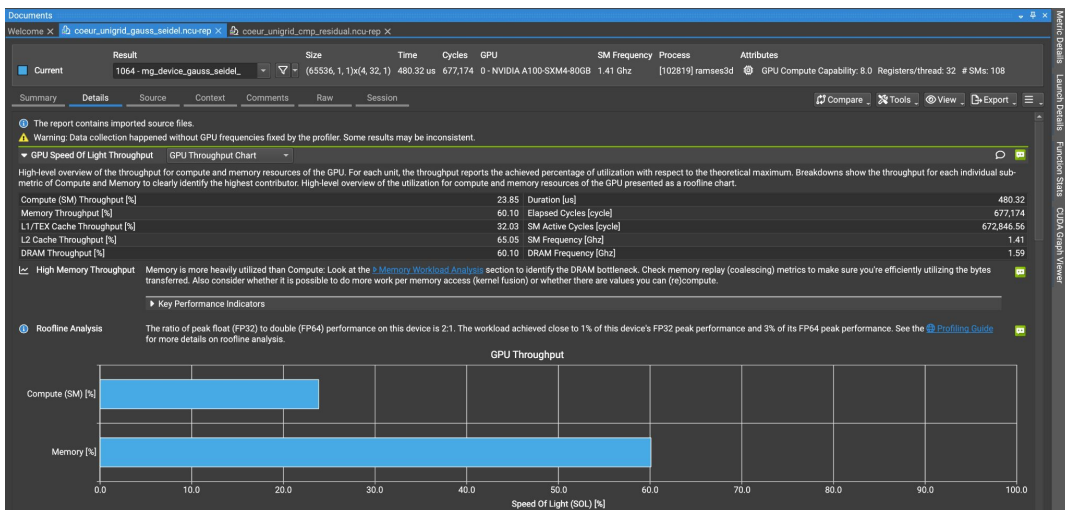


Progress and Goals

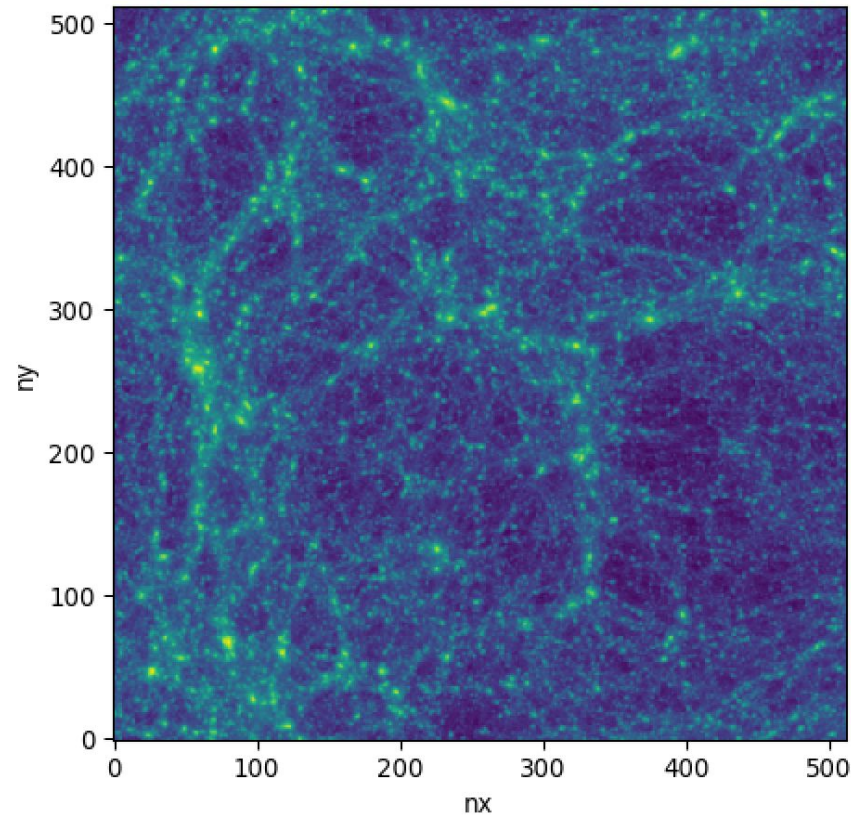
- We have profiled the code with NSight Systems
 - We have collected a few low hanging fruits
- We have benchmarked the gravity MG solver on A100 (H200) single precision
 - V-cycle: 2 (4) billion cell updates per GPU per second
 - MG hierarchy initialization: $\frac{1}{2}$ V-cycle
- Squeeze more performance: fuse more kernels, avoid D2H copies



Problems and Solutions



PARTICLE SOLVER



All together:
Isothermal hydro
+ DM + AMR

